4 Geometry

	Students should be taught to:		Notes
4.1 Angles, lines and triangles	**A**	distinguish between acute, obtuse, reflex and right angles	
	B	use angle properties of intersecting lines, parallel lines and angles on a straight line	Angles at a point, vertically opposite angles, alternate angles, corresponding angles, allied angles
	C	understand the exterior angle of a triangle property and the angle sum of a triangle property	
	D	understand the terms 'isosceles', 'equilateral' and 'right-angled triangles' and the angle properties of these triangles	
4.2 Polygons	**A**	recognise and give the names of polygons	To include parallelogram, rectangle, square, rhombus, trapezium, kite, pentagon, hexagon and octagon
	B	understand and use the term 'quadrilateral' and the angle sum property of quadrilaterals	The four angles of a quadrilateral are $90°$, $(x + 15)°$, $(x + 25)°$ and $(x + 35)°$ Find the value of x
	C	understand and use the properties of the parallelogram, rectangle, square, rhombus, trapezium and kite	
	D	understand the term 'regular polygon' and calculate interior and exterior angles of regular polygons	
	E	understand and use the angle sum of polygons	For a polygon with n sides, the sum of the interior angles is $(2n - 4)$ right angles
	F	understand congruence as meaning the same shape and size	
	G	understand that two or more polygons with the same shape and size are said to be congruent to each other	

	Students should be taught to:	Notes
4.3 Symmetry	**A** identify any lines of symmetry and the order of rotational symmetry of a given two-dimensional figure	Name a quadrilateral with no lines of symmetry and order of rotational symmetry of 2
4.4 Measures	**A** interpret scales on a range of measuring instruments	
	B calculate time intervals in terms of the 24-hour and the 12-hour clock	Use am and pm
	C make sensible estimates of a range of measures	
	D understand angle measure including three-figure bearings	
	E measure an angle to the nearest degree	
	F understand and use the relationship between average speed, distance and time	
	G use compound measure such as speed, density and pressure	Formula for pressure will be given
4.5 Construction	**A** measure and draw lines to the nearest millimetre	
	B construct triangles and other two-dimensional shapes using a combination of a ruler, a protractor and compasses	
	C solve problems using scale drawings	
	D use straight edge and compasses to: (i) construct the perpendicular bisector of a line segment (ii) construct the bisector of an angle	
4.6 Circle properties	**A** recognise the terms 'centre', 'radius', 'chord', 'diameter', 'circumference', 'tangent', 'arc', 'sector' and 'segment' of a circle	
	B understand chord and tangent properties of circles	Two tangents from a point to a circle are equal in length Tangents are perpendicular to the radius at the point of contact The line from the centre of a circle which is perpendicular to a chord bisects the chord (and the converse)

	Students should be taught to:	Notes
4.7 Geometrical reasoning	**A** give informal reasons, where required, when arriving at numerical solutions to geometrical problems	Reasons will only be required for geometrical calculations based on lines (including chords and tangents), triangles or polygons
4.8 Trigonometry and Pythagoras' theorem	**A** know, understand and use Pythagoras' theorem in two dimensions	
	B know, understand and use sine, cosine and tangent of acute angles to determine lengths and angles of a right-angled triangle	
	C apply trigonometrical methods to solve problems in two dimensions	To include bearings
4.9 Mensuration of 2D shapes	**A** convert measurements within the metric system to include linear and area units	e.g. cm^2 to m^2 and vice versa
	B find the perimeter of shapes made from triangles and rectangles	
	C find the area of simple shapes using the formulae for the areas of triangles and rectangles	
	D find the area of parallelograms and trapezia	
	E find circumferences and areas of circles using relevant formulae; find perimeters and areas of semicircles	
4.10 3D shapes and volume	**A** recognise and give the names of solids	To include cube, cuboid, prism, pyramid, cylinder, sphere and cone
	B understand the terms 'face', 'edge' and 'vertex' in the context of 3D solids	
	C find the surface area of simple shapes using the area formulae for triangles and rectangles	
	D find the surface area of a cylinder	
	E find the volume of prisms, including cuboids and cylinders, using an appropriate formula	
	F convert between units of volume within the metric system	e.g. cm^3 to m^3 and vice versa and 1 litre = 1000 cm^3

	Students should be taught to:	Notes
4.11 Similarity	**A** understand and use the geometrical properties that similar figures have corresponding lengths in the same ratio but corresponding angles remain unchanged	
	B use and interpret maps and scale drawings	